

Cedar Bluff Quarry Haul Road - Least-Cost Route Report (route CB-HR-01E)

Basis: drawing set CB-HR C-001 to C-004 Rev D and the two rasters it registers, with the three Rev E updates in the brief (plateau lease rate, Cedar Branch slip zone, G1 gate approach), which govern where they differ from the drawing set. Horizontal CRS WGS 84 / UTM zone 14N (EPSG:32614); lengths are horizontal. Status: route study only, not for construction.

1. Result

Quantity	Value
Total route cost, construction + loaded haulage (USD)	4,899,456.85
Construction cost (C-002 rates x grade factor)	4,709,285.66
Loaded haulage cost (C-004 D7: 8,000 USD/m x charged loaded rise)	190,171.20 (23.771 m of rise climbed loaded above 4.0 % effective)
Horizontal length, start cell centre to end cell centre (m)	6,062.40
Vertices (DEM cell centres) / moves / deflection vertices	507 / 506 / 34
Start cell centre (T1 on CR-114)	E 583155.00 N 3349105.00 z 374.54 m
End cell centre (G1 re-surveyed, C-004 Table 6)	E 586455.00 N 3350105.00 z 484.97 m
Creek crossing	Approved window X-2, culvert on 2 watercourse cells at N 3349555
Crossing through the other window (X-1)	None compliant: no route exists through X-1 (Section 5)
Max grade rising / falling in chainage direction (T1 10.0 %)	+10.00 % / -5.73 %
Max loaded uphill grade (D3: 9.0 - 3.0 = 6.0 %)	5.73 %
Longest continuous run of moves > 8.0 % (T3 60.0 m)	56.6 m
Longest run > 8.0 % touching the slip zone N 3,349,600 to 3,349,800 (limit 30.0 m)	28.3 m
Route length on plateau lease (grassland at or above 1,500.00 ft, 1,300 USD/m)	1,826.7 m
G1 gate approach (last 40.0 m straight, due west)	4 westward moves from E 586495 to E 586455 at N 3350105
Largest change of direction between moves (T4 45 deg)	45 deg
Smallest K between consecutive moves (D5 1.4 m/%)	1.404 m/%
Shortest tangent between deflection vertices (D6: $\geq 2R \tan 22.5 = 37.28$ m)	40.00 m
Start to first / last to end deflection (D6: $\geq R \tan 22.5 = 18.64$ m)	30.00 m / 40.00 m
Escarpment ascent	northern slump; route reaches N 3351495 at E 586015, then returns south along the plateau to G1
Formation edge to wetland W-1 / to HS-1	675.8 m (limit 30.0) / 446.6 m (limit 150.0)

2. Method

Terrain. DEM DN converted with $\text{Elevation(ft)} = 1150.00 + 0.01 \times \text{DN}$ (C-002 Table 1), then to metres with the US survey foot (1200/3937 m). Grades use metres on both axes.

Georeferencing. Both rasters are pixel-is-area with the registered upper-left corner. DEM cell centres are at E 583 005 + 10i, N 3 351 495 - 10j. The land-cover raster has its own origin (E 582 900, N 3 351 620) and 20 m pixels; each DEM cell takes the class of the land-cover pixel containing its centre (exact RGB match).

Control points. T1, X-1, X-2 and HS-1 from C-002 Table 3 (WGS 84) with PROJ. G1 from C-004 Table 6: TxDOT surface coordinates E 3,001,506.17 ft, N 10,073,679.76 ft, divided by the surface adjustment factor 1.00012 to grid E 3,001,146.03 ft, N 10,072,471.06 ft in NAD83 / Texas Central, US survey feet (EPSG:2277), then transformed with PROJ to E 586,454.90, N 3,350,102.50 (cell E 586455, N 3350105). Using the surface values as grid puts G1 about 110 m east and 368 m north; reading the feet as international feet moves it about 6 m south into the next cell; the superseded Rev C position (C-002 Table 3, still plotted on C-001) is two cells west.

Road footprint (C-003). Formation $1.00 + 1.00 + 10.00 + 1.00 + 1.00 = 14.00$ m, 7.00 m either side of the centreline; setback and exclusion measured to the formation edge along every move (centreline clearance > 37.0 m from wetland pixels, > 157.0 m from HS-1).

C-003 Table 4. T1 every move at most 10.0 %; T2/T3 consecutive moves steeper than 8.0 % total at most 60.0 m; T4 direction changes by at most 45 deg between consecutive moves.

C-004 loaded direction (D1-D3). Loaded trucks run G1 to T1, against chainage. A move that falls in chainage direction is climbed loaded; its effective grade is grade + 3.0 % rolling resistance, limited to 9.0 %, so its grade may not exceed 6.0 %. Moves rising in chainage direction are loaded downhill and keep the 10.0 % T1 limit.

Vertical curvature (D5). For consecutive moves of lengths L1, L2 and signed grades g1, g2, the distance between their midpoints along the centreline is $(L1 + L2)/2$, so $|g2 - g1| \leq (L1 + L2) / (2 \times 1.4)$: 7.14 % between two orthogonal moves, 8.62 % orthogonal-diagonal, 10.10 % between two diagonal moves.

Horizontal curves (D6). Every deflection on the grid is 45 deg, so an arc of radius 45 m needs a tangent length $T = R \tan(22.5 \text{ deg}) = 18.64$ m on each side. Arcs may not overlap, so successive deflection vertices must be at least $2T = 37.28$ m apart (4 orthogonal or 3 diagonal moves), and the first and last deflection vertices at least $T = 18.64$ m from the start and end cell centres (2 moves).

Objective (C-004 Note D1, D7). Move cost = $d \times (c_i + c_j)/2 \times f(|g|) + 8,000 \times (z_i - z_j)$ when the move is climbed loaded (falls in chainage direction) at an effective grade above 4.0 %, i.e. a grade above 1.0 % once the 3.0 % rolling resistance is added; otherwise the construction term alone. The fall is not credited; the 450 USD/m brake-wear and 6.0 % empty top-gear figures in Table 7 are information only.

Rev E (a) plateau lease. A DEM cell whose class is Grassland / pasture and whose ground elevation is at or above 1,500.00 ft takes a base unit rate of 1,300 USD/m instead of 420 USD/m. The test is made on the DEM value in feet $(1150.00 + 0.01 \times \text{DN})$, i.e. 457.201 m; no DEM cell lies exactly on the threshold.

Rev E (b) Cedar Branch slip zone. Any continuous run of moves steeper than 8.0 % that contains a move with either end cell centre between N 3,349,600 and N 3,349,800 is limited to 30.0 m over its whole length, including the part outside the band. The steep-run state carries a flag set by the first such move and cleared when the run ends.

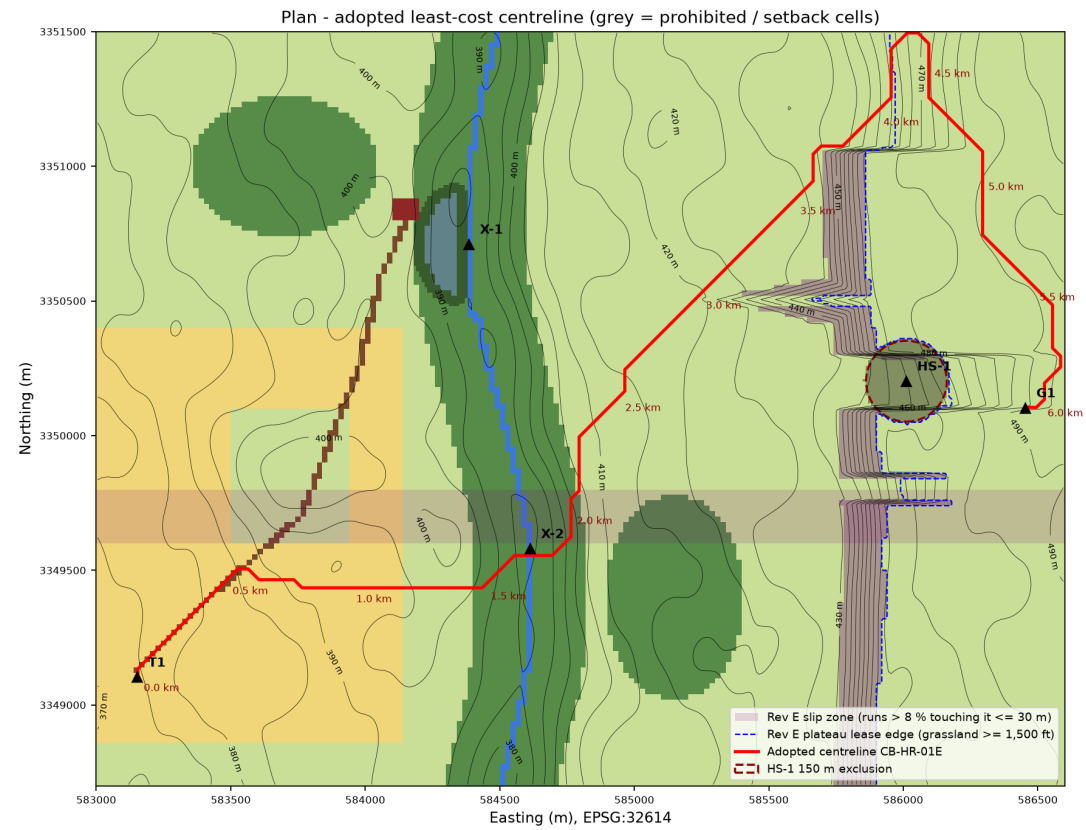
Rev E (c) gate approach. The last four moves into the G1 cell are due west (40.0 m straight), so the route enters the gate cell from the east side. Every drawing-set rule applies on the approach as elsewhere.

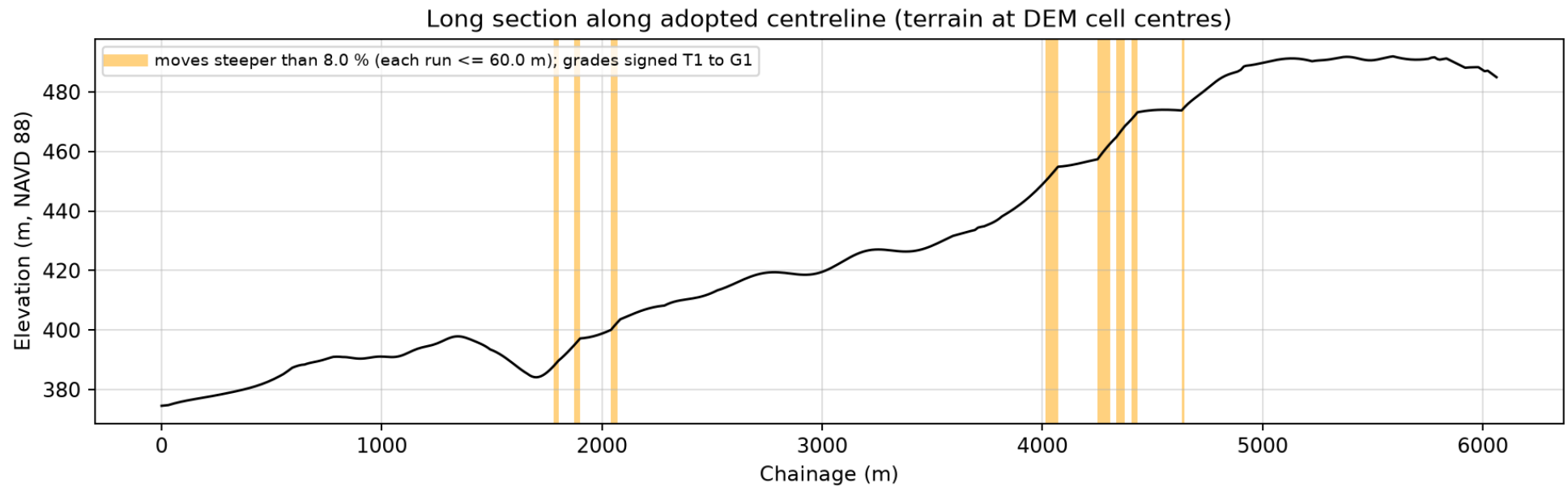
Search. T3, T4, D5 and D6 depend on the path, so the search runs on an expanded state: cell x arrival heading (which also fixes the previous move and its grade for D5) x steep-run length and slip-zone flag (for T3 and Rev E b) x moves since the last deflection vertex, with a separate counter for the first tangent (for D6). The gate approach removes every move into the last four approach cells other than the westward one. Dijkstra on this graph gives the exact optimum; the end state is accepted only if the last tangent is at least 18.64 m.

Rates and constraint values applied (C-002, C-003, C-004 Rev D)

Item applied	Value
Grassland / pasture; cultivated cropland	420 / 480 USD/m
Grassland / pasture at or above 1,500.00 ft (Rev E plateau lease)	1,300 USD/m
Rev E slip zone; gate approach	N 3,349,600 to 3,349,800, runs > 8.0 % touching it <= 30.0 m; last 40.0 m due west into G1
Woodland (Rev B; Rev A 690 withdrawn); rock outcrop	940 / 1,150 USD/m
Existing gravel track; culvert cell within 30.0 m of X-1 / X-2	160 / 3,800 USD/m
Watercourse elsewhere, wetland, farmstead	prohibited
Formation width; wetland setback; HS-1 radius (to formation edge)	14.0 m; 30.0 m; 150.0 m
T1 max grade; T2/T3 steep threshold / run; T4 turn	10.0 %; > 8.0 % / 60.0 m; 45 deg
D1 loaded direction; D2 rolling resistance; D3 effective limit	G1 to T1; 3.0 %; 9.0 % (grade 6.0 %)
D5 minimum K; D6 minimum radius	1.4 m per %; 45.0 m (tangents 37.28 m / 18.64 m)
D7 loaded haulage	8,000 USD per m of loaded rise on moves above 4.0 % effective (1.0 % grade)
Grade factor f(g)	1.00 (g<=3); 1.00+0.05(g-3) (3<g<=6); 1.15+0.12(g-6) (6<g<=10)

3. Plan and long section





4. Breakdown

Land-cover class	Length (m)	Construction cost (USD)
Grassland / pasture	4,260.2	3,611,622
Cultivated cropland	624.0	300,163
Woodland	584.9	628,117
Existing gravel track	573.3	92,731
Watercourse at X-2 (culvert)	20.0	76,654
Total	6,062.4	4,709,286

Each move's length is split equally between its two end cells; its construction cost is split in proportion to the two cells' unit rates. Loaded haulage is not allocated to classes.

Grade band (abs)	0-3 %	3-6 %	6-8 %	8-10 %
Length (m)	3,949.2	1,489.9	351.1	272.1

5. What controls the alignment

Creek crossing - X-1 is not feasible. At X-1 the eastern watercourse column is 35.0 m from the W-1 boundary; the formation edge would be $35.0 - 7.0 = 28.0$ m from the wetland. With X-2 removed the search finds no compliant route, so the only crossing is X-2.

Cedar Bluff escarpment. The saddle gap next to HS-1 is closed by the 150 m exclusion measured to the formation edge; the spur crest grades at about 9.2 % for over 600 m and breaks T3 (60 m); the chute needs 90 and 135 deg switchbacks and breaks T4. The route climbs the northern slump and returns south along the plateau to G1.

C-004. The loaded-uphill limit (6.0 % on moves falling in chainage direction) reshapes the descent into the Cedar Branch valley and the plateau return; applied the wrong way round (to chainage-rising moves) or to both directions it closes every escarpment ascent, so no route exists. The 45 m radius forces tangents of at least 37.28 m between deflections and removes the short zig-zags the unconstrained optimum uses. $K = 1.4$ alters the slump climb where steep runs are broken. Loaded haulage adds 190,171 USD and moves the route to one with less loaded climbing above the top-gear range; because of the 4.0 % threshold, charging the rise in chainage direction, dropping the rolling resistance from the threshold or charging every loaded rise each gives a different line. Each run below breaks exactly one stated rule and is not an option.

Rev E. The plateau lease cannot be avoided (G1 is on the plateau). At 1,300 USD/m the grade factor multiplies a larger base rate, which changes the trade-off between grade, length and loaded haulage on the plateau: the plateau leg moves off the line it takes without the lease, although the length on leased ground (1,826.7 m) is the same. The slip zone breaks the two 56.6 m steep runs of the Rev D climb out of the Cedar Branch valley into runs of at most 28.3 m. The gate approach turns the end of the route to arrive from the east. Applying the 30 m limit only to moves inside the band, or ignoring any one of the three updates, gives a different line.

Run	Total (USD)	Length (m)	Vertices	Xing	Max loaded uphill %	Min K	Min tangent (m)
Adopted route (drawing set Rev D + Rev E updates)	4,899,457	6,062.4	507	X-2	5.73	1.40	40.0
Check: Rev E updates ignored (Rev D basis)	3,110,904	5,974.5	492	X-2	5.88	1.43	40.0
Check: plateau lease rate not applied	3,158,560	6,028.3	504	X-2	5.73	1.43	40.0
Check: lease rate applied to all grassland	6,898,660	5,963.8	493	X-2	5.88	1.40	40.0
Check: slip-zone 30 m limit tested only on moves inside the band	4,891,903	6,033.1	502	X-2	5.73	1.40	40.0
Check: slip-zone limit ignored	4,859,069	6,032.1	499	X-2	5.88	1.40	40.0
Check: gate approach ignored	4,871,082	6,039.0	503	X-2	5.73	1.40	40.0
Check: gate approach of 3 moves (30 m)	4,890,415	6,056.5	506	X-2	5.73	1.40	40.0
Check: D3 loaded adverse limit ignored (or 9.0 % read as grade)	4,866,975	6,029.7	500	X-2	8.41	1.40	40.0
Check: D5 vertical curvature ignored	4,773,748	6,071.7	510	X-2	5.73	1.19	40.0
Check: D6 curve radius ignored	4,784,702	6,069.1	494	X-2	5.55	1.40	10.0
Check: D6 with $R \tan(22.5)$ between deflections	4,806,473	6,086.7	497	X-2	5.55	1.40	20.0
Check: D7 haulage ignored (construction cost only)	4,683,990	6,121.8	503	X-2	5.55	1.43	40.0
Check: D7 rise charged in chainage direction	5,752,017	6,087.2	507	X-2	5.73	1.40	40.0
Check: D7 threshold 4.0 % read as grade (no rolling resistance)	4,762,100	6,146.7	503	X-2	5.55	1.43	40.0

Check: D7 threshold ignored (every loaded rise charged)	4,918,901	6,062.4	507	X-2	5.73	1.40	40.0
Check: G1 at superseded Rev C position	4,423,038	5,744.7	486	X-2	5.73	1.43	40.0
Check: EPSG:2277 read as international feet	5,407,733	6,465.5	544	X-2	5.73	1.40	40.0
Check: C-004 ignored entirely (Rev C rules)	3,910,971	5,590.0	461	X-2	8.14	1.19	10.0
Check: 45 deg direction-change limit (T4) ignored	4,598,997	5,690.3	476	X-2	5.73	1.47	40.0
Check: 60 m sustained-grade limit (T3) ignored	3,821,690	4,644.8	405	X-2	5.73	1.75	40.0
Check: without HS-1 exclusion	2,948,058	4,167.8	366	X-2	5.73	1.55	40.0
D3 applied to chainage-rising moves / to both directions	no route						
D5 read as a flat 7.14 % grade-change limit	no route						
Only X-1 available	no route						
Gate approach read as travelling east into G1	no route						
G1 surface coordinates used as grid (SAF ignored): gate approach leaves the DEM	no route						

6. Compliance and limitations

Every move is at or below 10.0 % (max 10.00 %); loaded uphill at most 5.73 %; the longest run over 8.0 % is 56.6 m; no change of direction exceeds 45 deg; $K \geq 1.404$; tangents ≥ 40.00 m (ends 30.00 / 40.00 m); the formation edge stays 675.8 m from W-1 and 446.6 m from HS-1; the only watercourse cells used are within 30 m of X-2; the longest steep run touching the slip zone is 28.3 m; the last 40.0 m run due west into G1.

The centreline is a 10 m raster least-cost path screened against C-004 geometry; it is a corridor, not a geometric design. Tie-break order, international versus US survey foot for the DEM, and thresholds moved by 0.02 (grade), 0.01 (K) or 1 m (radius) give the same route.

Deliverables: CB_HaulRoad_Centreline.gpkg (layer 'centreline', one LineString, EPSG:32614, fields route_id, total_cost_usd, length_m), CB_HaulRoad_Vertices.csv (one row per vertex, start to end, 11 columns; unit_rate_usd_per_m is the base rate of the vertex cell; cum_cost_usd = cum_constr_usd + cum_haul_usd), this report.